



Mirta Galesic

Collective Adaptation and Belief dynamics

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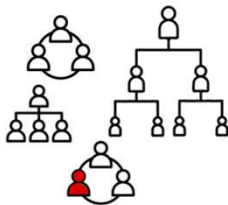
1

Collective adaptation

Families

Budget
decisions?Next
vacation?

Organizations



Social movements



Political systems

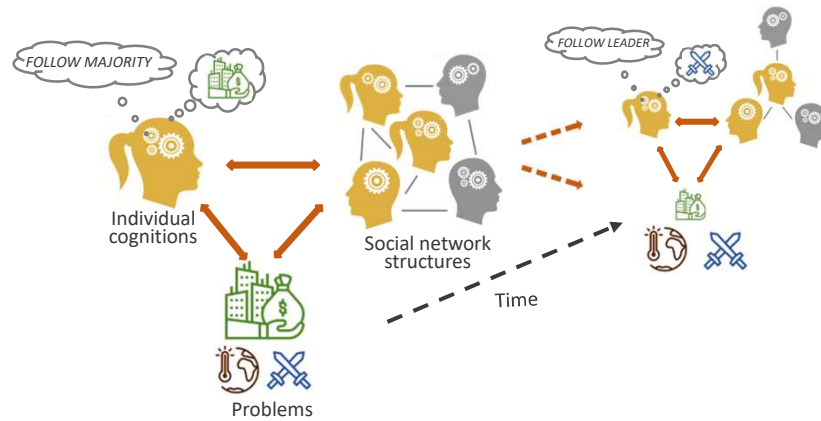


Why is collective adaptation to
some problems so difficult?

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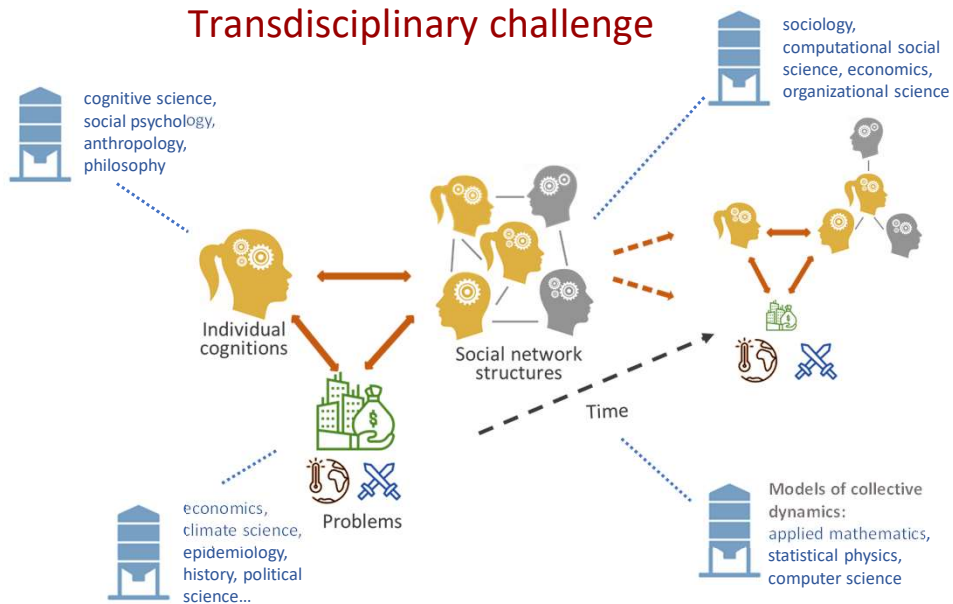
Collective adaptation

Complex system of individual cognitions, social networks, and problems we face



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Transdisciplinary challenge



4

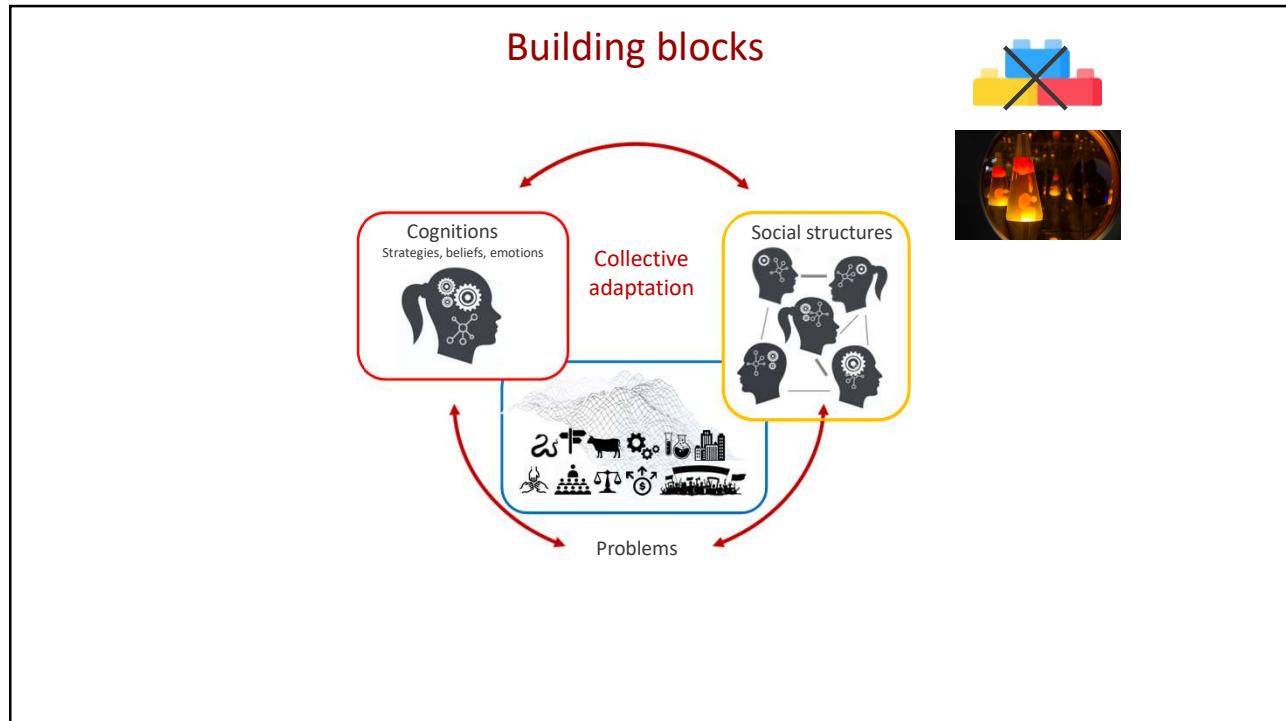
Overlapping views, different labels

- Collective intelligence (Graves, 1842, Woolley et al, 2010, Malone & Bernstein, 2015)
- Social learning (Bandura, 1977; Hoppitt & Laland, 2013; Kendal et al, 2018; Yaniv, 2004)
- Collective problem solving (Conradt & Roper, 2010; Gonzalez et al., 2015; Hills et al., 2015; Mehlhorn et al., 2015)
- Wisdom of crowds (Condorcet, 1783; Galton, 1907; Davis-Stober et al., 2014; Mellers et al., 2014; Budescu & Chen, 2015)
- Group decision making (Page, 2008; Stasser & Titus, 1985)
- Belief dynamics (Centola & Macy, 2007; Epstein, 2014; Galesic et al., 2021; Pentland, 2014; Proskurnikov & Tempo, 2017; Vallacher, Read, & Nowak, 2017)
- Cultural evolution (Boyd & Richerson, 1985; Mesoudi, 2016)
- Game theory (Friedman, 1998; Newton, 2018; Ostrom, 2010)
- Group minds (Goldstone & Theiner, 2017; Hinsz et al., 1997)
- ...

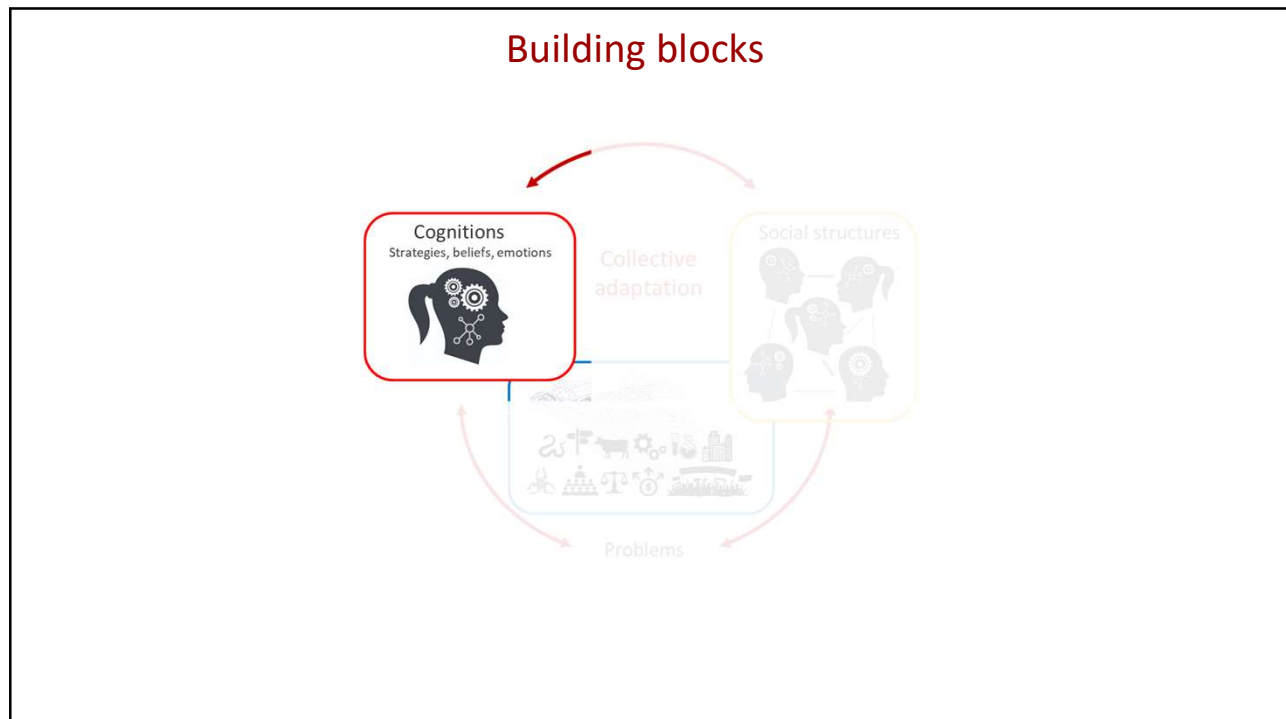
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Building blocks of collective adaptation

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Social cognitive strategies

Strategies for social interactions (Hertwig & Hoffrage, 2013):

- **integration of social information** (Hoppit & Laland, 2013)
- coordination (Grice, 1975; Moussaid et al., 2011)
- cooperation (Axelrod, 1984; Bowles & Gintis, 2013)
- exploration (Hills, et al., 2015; Mehlhorn et al., 2015)
- network building and revision (Jackson, 2010)
- innovation, etc.



Studied under different labels:

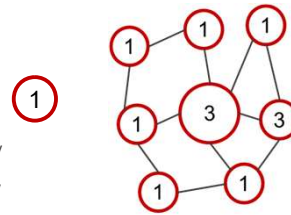
- Social learning strategies
- Belief updating strategies
- Group decision-making rules
- Voting procedures
- Aggregation procedures

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Strategies for integrating social information

Three basic classes:

- **Frequency-dependent strategies: majority, plurality, unanimity, minority rules, complex contagion**
Condorcet 1785; political science (Heinberg, 1932); statistics (Penrose, 1946); psychology (Asch, 1955); economics (Plott, 1967); cultural evolution (Boyd & Richerson, 1985); computer science (Parhami, 1994); statistical physics (Krapivsky & Redner, 2003); biology (King & Cowlishaw, 2007); sociology (complex contagion; Centola & Macy, 2007)
- **Averaging strategies: with or without weights, e.g. advice taking, blending inheritance, voter model, contagion rules**
Galton, 1907; economics (DeGroot, 1974; Golub & Jackson, 2010); advice taking (Molleman et al., 2020; Yaniv, 2004); statistical physics (Ising models; Castellano et al., 2009); cultural evolution (blending inheritance: Boyd & Richerson, 1985); network science (contagion; Newman, 2003)
- **Model-based strategies: follow leader, expert, similar, confident, liked, best**
cultural evolution (high status: Henrich & Gil-White, 2001), social psychology (liking, authority: Cialdini & Trost, 1998), cognitive psychology (similarity, Wisdom et al., 2013), law (confidence; Penrod & Cutler, 1995)



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Different beliefs about important problems



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Group emotions

Current Directions in Psychological Science

aps ADVANCING PSYCHOLOGICAL SCIENCE Journal indexing and metrics

Restricted access | Research article | First published online February 24, 2020

Collective Emotions

Amit Goldenberg, David Garcia, et al., and James J. Gross [View all authors and affiliations](#)

Volume 29, Issue 2 | <https://doi.org/10.1177/0963721420901574>

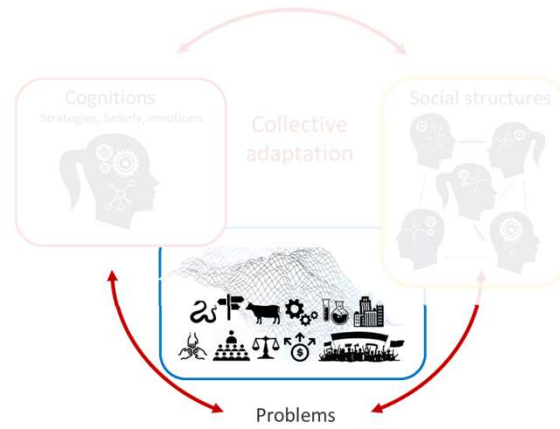
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Abstract

When analyzing situations in which multiple people are experiencing emotions together—whether the emotions are positive or negative and whether the situations are online or offline—we are intuitively drawn to the emotions of each individual in the situation. However, this type of analysis often seems incomplete. In

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Building blocks



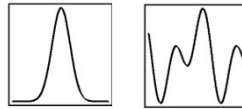
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Problems

– Categorical or continuous



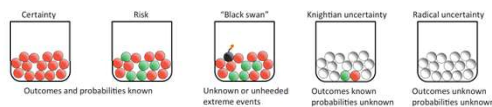
– Simple or complex



– One-shot or repeated problems



– Layers of uncertainty



– Rate and type of change



– Structure of incentives



Meder et al., 2013, TICS

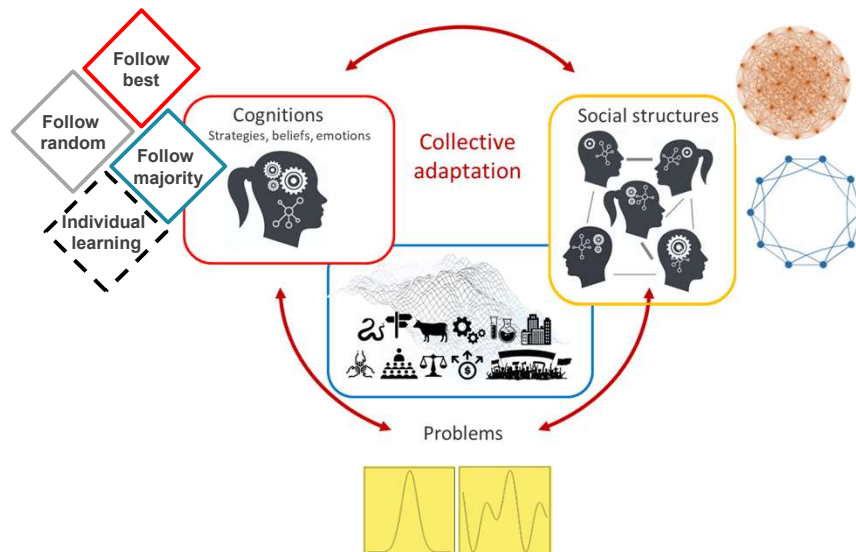
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Interaction of building blocks: an example

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Interaction of building blocks: an example

Barkoczi & Galesic (2016), *Nature Communications*

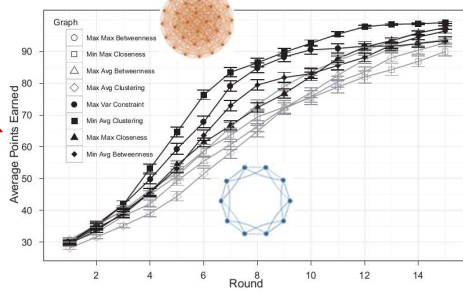
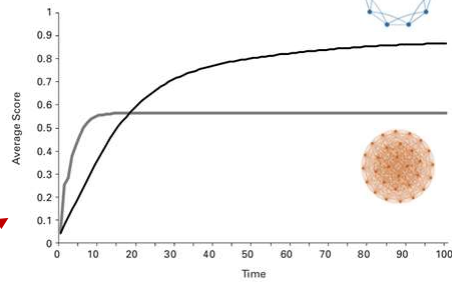


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Interaction of building blocks: an example

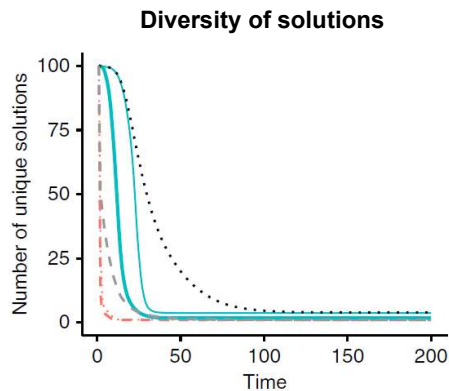
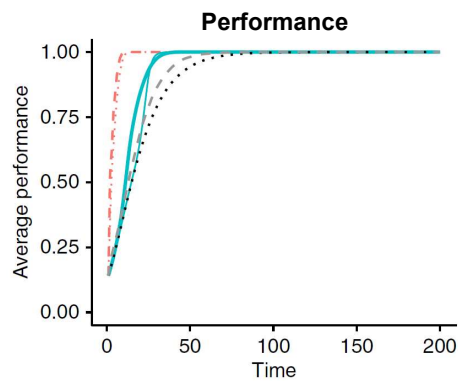
Contradictory findings about network structure and wisdom of crowds

- Poorly-connected, slow networks better (Lazer & Friedman, 2007, ASQ; Derex & Boyd, 2016, PNAS)
- Well-connected, fast networks better (Mason & Watts, 2012, PNAS)



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Simple tasks: fast learning strategies better

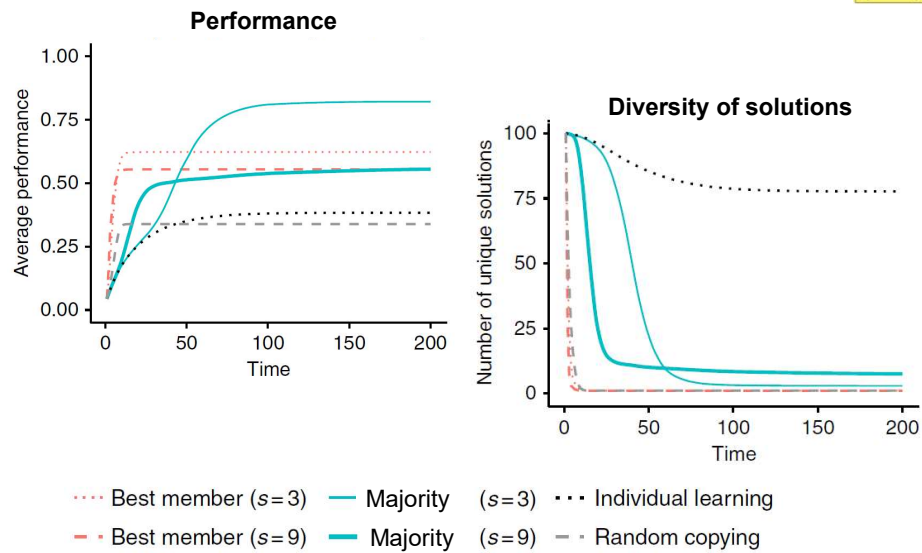
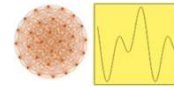


..... Best member ($s=3$) — Majority ($s=3$) Individual learning
 - - - Best member ($s=9$) — Majority ($s=9$) - - Random copying

Barkoczi & Galesic (2016), *Nature Communications*

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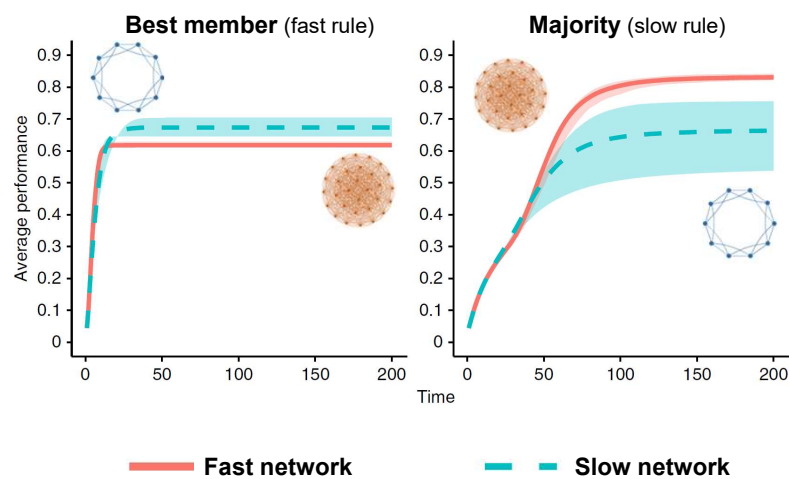
Complex tasks: slow learning strategies better



Barkoczi & Galesic (2016), *Nature Communications*

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Interplay of networks and learning strategies



Barkoczi & Galesic (2016), *Nature Communications*

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Interaction of building blocks: an example

Contradictory findings about network structure and wisdom of crowds

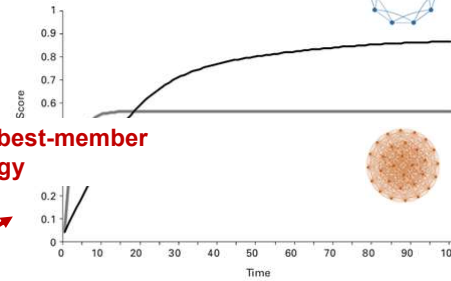
- Poorly-connected, slow networks better (Lazer &

Apparent contradictions can be resolved by taking into account the whole system of cognition and networks

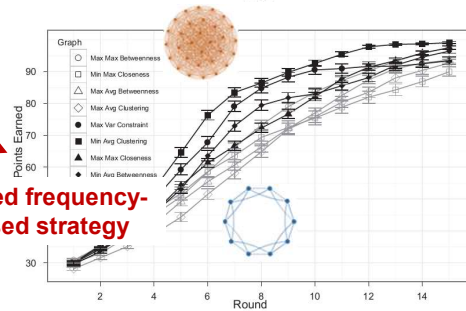
- Well-connected, fast networks better (Mason & Watts, 2012, PNAS)



Used best-member strategy



Used frequency-based strategy

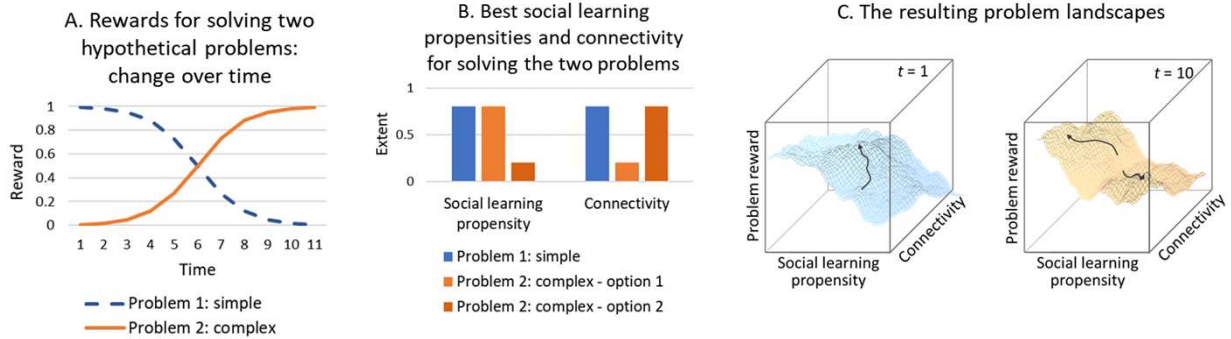


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Emergence of collective adaptation

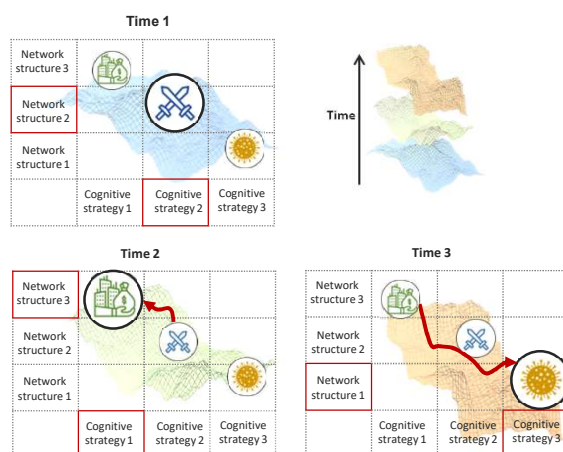
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A simple example



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Emergence of collective adaptation



Implications

- Path dependence: past adaptations affect future ones
- Satisficing on multiple tasks instead of optimizing for one task
- Collective myopia instead of willful resistance to change

Galesic et al. (2023). Beyond collective intelligence: Collective adaptation. *Journal of the Royal Society Interface*.

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Quantitative models of collective adaptation

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Modeling challenges

Models of collective adaptation ...

- Should be simple, but grounded in theories of human cognition and sociality
- Should represent the dynamic interplay of cognition, network, and problem structures
- Should produce quantitative predictions that can be tested by empirical data

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Modeling challenges

SCIENCE
4 August 1972, Volume 177, Number 4047

More Is Different

P. W. Anderson

seems to me that one may array the sciences roughly linearly in a hierarchy, according to the idea: The elementary entities of science X obey the laws of science Y.

X	Y
solid state or many-body physics	elementary particle physics
chemistry	many-body physics
molecular biology	chemistry
cell biology	molecular biology
•	•
•	•
•	•
psychology	physiology
social sciences	psychology

But this hierarchy does not imply that science X is "just applied Y." At each stage entirely new laws, concepts, and generalizations are necessary, requiring inspiration and creativity to just as great a degree as in the previous one. Psychology is not applied biology, nor is biology applied chemistry.

Trends in Ecology & Evolution

CellPress
REVIEWS

Opinion

A User's Guide to Metaphors In Ecology and Evolution

Mark E. Olson,^{1,*} Alfonso Arroyo-Santos,² and Francisco Vergara-Silva³

Biologists energetically debate terminology in ecology and evolution, but rarely discuss general strategies for resolving these debates. We suggest focusing on metaphors, arguing that, rather than looking down on metaphors, biologists should embrace these terms as the powerful tools they are. Like any powerful

Highlights
Biologists constantly debate specific terms in ecology, but rarely discuss general strategies for resolving these debates. Most select biological terms.

Psychological Review
1981, Vol. 18, No. 2, 204-207

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0033-2909/81/0000-0000\$01.00/0

From Tools to Theories: A Heuristic of Discovery in Cognitive Psychology

Gerd Gigerenzer
Center for Advanced Study in the Behavioral Sciences
Stanford, California

The study of scientific discovery—where do new ideas come from?—has long been designated by philosophers as irrelevant to analyzing the growth of scientific knowledge. In particular, little is known about how cognitive theories are discovered, and neither the classical accounts of discovery

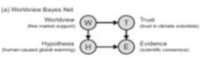
WARNING: Physics Envy May Be Hazardous To Your Wealth!*

Andrew W. Lo[†] and Mark T. Mueller[‡]

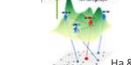
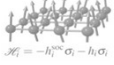
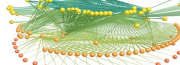
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Modeling collective adaptation

Models from cognitive/social sciences

Reinforcement learning	Fast and frugal heuristics	Bayesian reasoning	Expected utility models
 Wolpert & Turner, 2001	 Gigerenzer & ABC group	 Cook & Lewandowsky, 2016	Expected utility models Social learning models Models of institutions ...

Models from other sciences

Evolutionary biology	Statistical physics	Ecology
- Adaptive landscapes - Cultural evolution - Multi-level selection - Adaptive dynamics  Ha & Jeong, 2022	- Ferromagnetic models - Random walk models  Tump et al., 2022	Dynamic interaction networks  Jen Dunne

Combinations of analogies

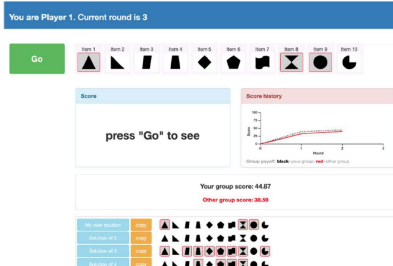
Different analogies for different building blocks of collective adaptation, for example:

- Drift diffusion model of learning + an evolutionary process (Tump et al., 2022)
- Epidemiological + evolutionary models to study evolution on sociality on different time scales (Cooney et al., 2022)

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Testing and refining the models

- Group experiments



- Surveys



- Studies of real-world collectives

- Activist groups
- Sustainable food initiatives



- LLM-based agent-based models

- LLMs can mimic different “types” of people
- Or even actual people
- **With caution!**



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Belief dynamics

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Different beliefs about important problems within collectives

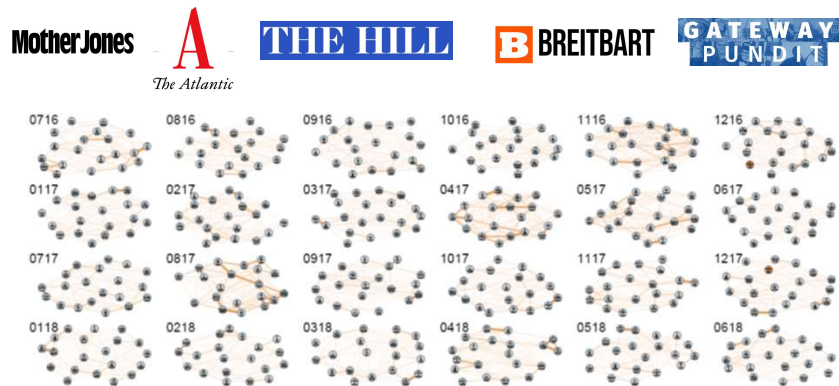


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Different beliefs about important problems

→ Different belief networks

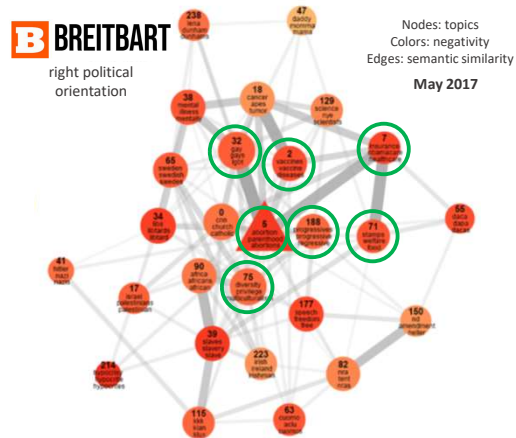
Illustration: Beliefs expressed in comment sections of U.S. news sites across the political spectrum, extracted using BERTopic and sentiment analysis, for each month over 5 to 8 years, from 401 million comments posted by 643 thousand commenters (Bacaksizlar Turbic & Galesic, 2023, *Scientific Reports*; Jaksic et al., in prep)



Ha, Jaksic, Rodriguez, Pellert, Olsson, Bacaksizlar Turbic, & Galesic, in prep

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→ Different belief networks: **Abortion**



Models of belief dynamics

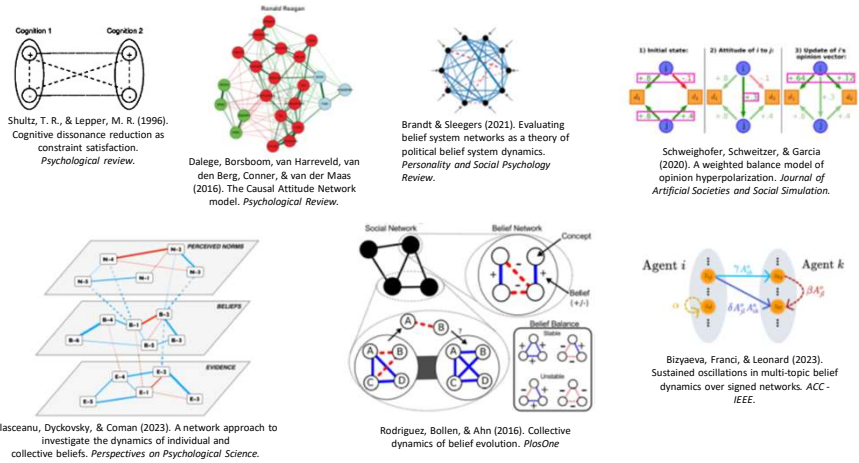
Many analogies for modeling belief dynamics

- **Epidemiological models**, where transmission of belief is like a transmission of disease (Newman, 2003; Cooney et al., 2022)
- **Percolation**, where beliefs seep through a society like liquid through a substance (Duffie et al., 2010; Li & Wang, 2019)
- **Ferromagnetic models**, where beliefs spread in a society like a crystal lattice (Duffie et al., 2010; Li & Wang, 2019)
- **Balance**, where beliefs spread in a society like a network of consistent relationships on the level of pairs and triads (Heider, 1958; Pham et al., 2020)
- **Expected utility**, where beliefs change in line with a weighted average of different cognitions (Ajzen, 1991; Friedkin & Bullo, 2017)
- **Evolution**, where beliefs evolve in the process of cultural learning (Richerson & Boyd, 2008; Anderson & Creanza, 2022)
- **Neural networks**, where beliefs change to satisfy constraints (Monroe & Read, 2008; Shultz & Lepper, 1996; Van Overwalle & Siebler, 2005)
- **Bayesian networks**, where networks of beliefs change in line with their conditional dependencies (Cook & Lewandowsky, 2016; Pallavicini, 2021)
- ...

Galesic, Olsson, Dalege, van der Does, & Stein (2021). Integrating social and cognitive aspects of belief dynamics: towards a unifying framework. *Journal of the Royal Society Interface*.

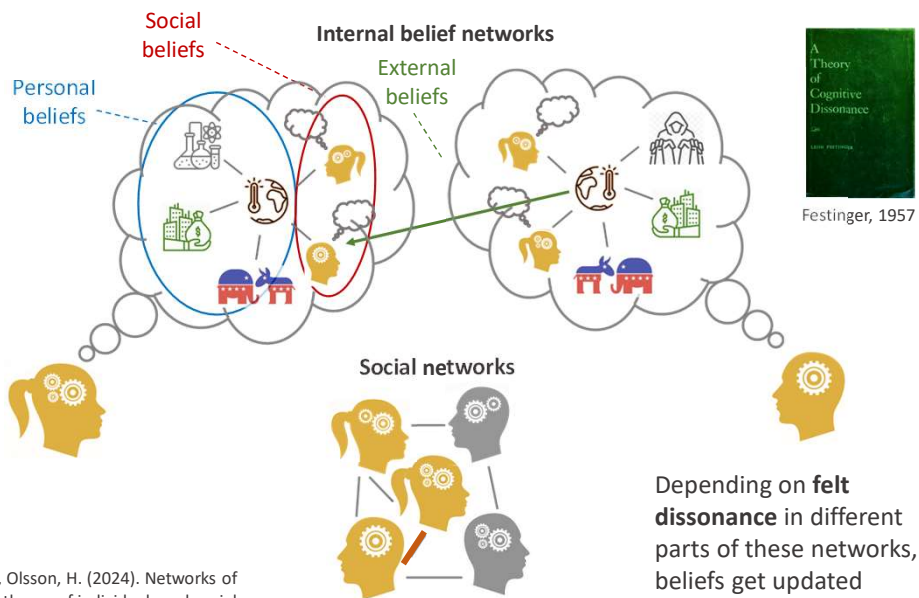
Olsson & Galesic (2024) Analogies for modeling belief dynamics. *Trends in Cognitive Science*

Models of belief dynamics: Belief networks



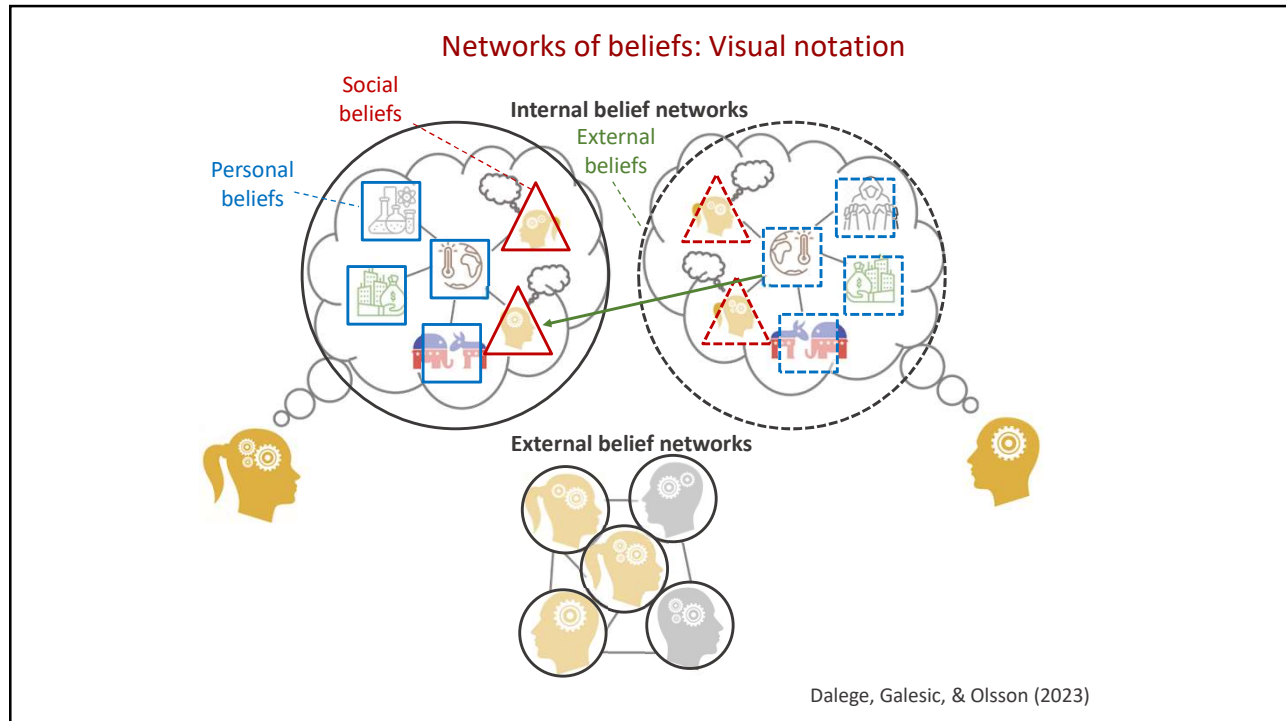
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Integrating social and belief networks

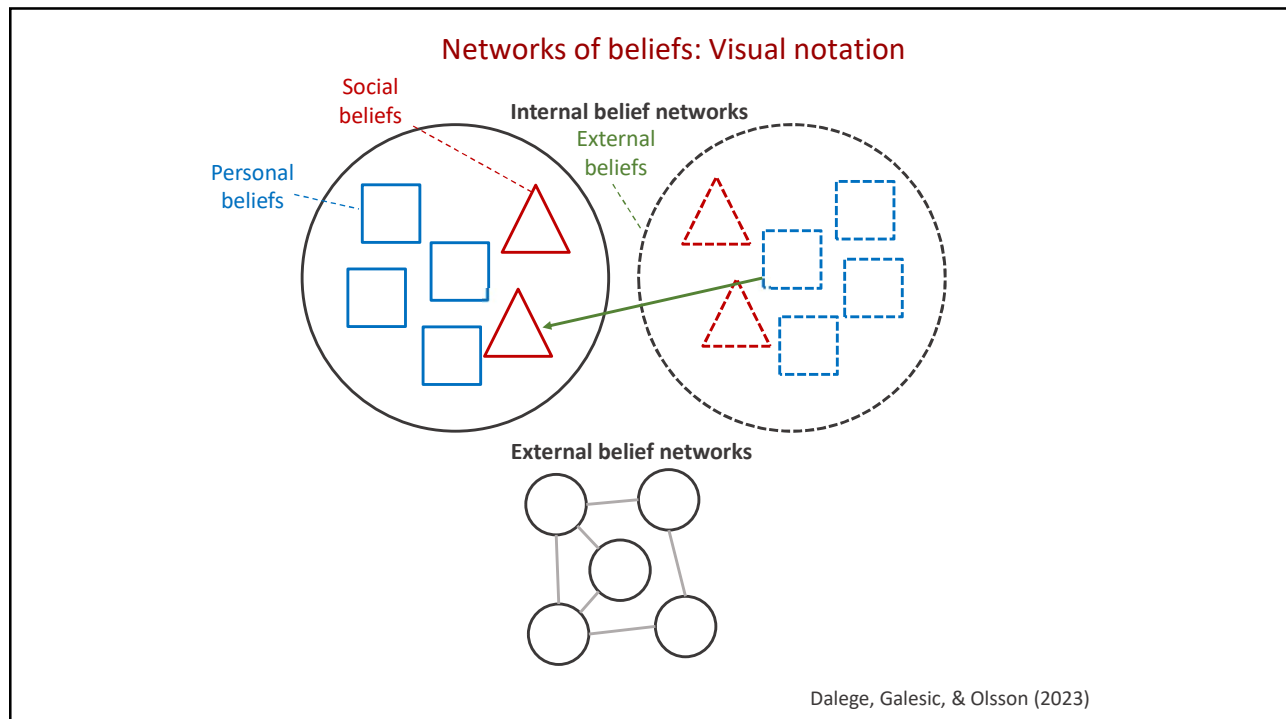


Dalege, J., Galesic, M., Olsson, H. (2024). Networks of beliefs: An integrative theory of individual- and social-level belief dynamics. *Psychological Review*

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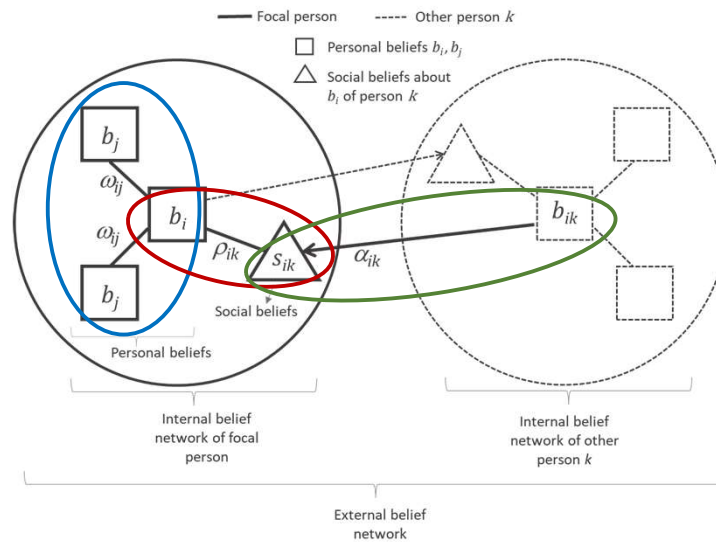


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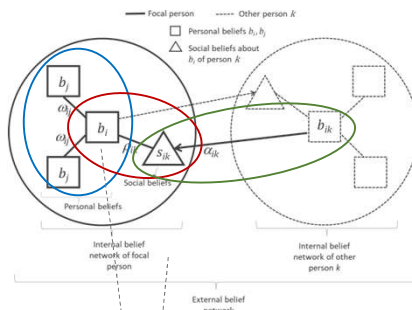
Networks of beliefs: Mathematical notation



Dalege, Galesic, & Olsson (2023)

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Networks of beliefs: Mathematical notation



Updating personal beliefs

$$P(b_i \rightarrow b'_i) \approx 1/1 + e^{\Delta(\beta_{pers}H_{pers} + \beta_{soc}H_{soc})}$$

Updating social beliefs

$$P(s_i \rightarrow s'_i) \approx 1/1 + e^{\Delta(\beta_{soc}H_{soc} + \beta_{ext}H_{ext})}$$

Potential dissonances

Personal dissonance

$$H_{pers} = -\sum_{ij} \omega_{ij} b_i b_j$$

Social dissonance

$$H_{soc} = -\sum \rho_{ik} b_i s_{ik}$$

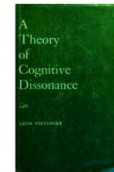
External dissonance

$$H_{ext} = -\sum_i \sum_k \alpha_k s_{ik} b_{ik}$$

Felt dissonances

Depend on **attention** to personal, social, and external dissonances

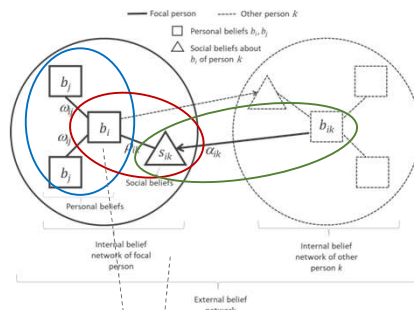
$$\beta_{pers}, \beta_{soc}, \beta_{ext}$$



Dalege, Galesic, & Olsson (2023)

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Networks of beliefs



Updating personal beliefs

$$P(b_i \rightarrow b'_i) \approx 1/1 + e^{\Delta(\beta_{pers}H_{pers} + \beta_{soc}H_{soc})}$$

Updating social beliefs

$$P(s_i \rightarrow s'_i) \approx 1/1 + e^{\Delta(\beta_{soc}H_{soc} + \beta_{ext}H_{ext})}$$

Potential dissonances

Personal dissonance

$$H_{pers} = - \sum_{ij} \omega_{ij} b_i b_j$$

Social dissonance

$$H_{soc} = - \sum_{ik} \rho_{ik} b_i s_{ik}$$

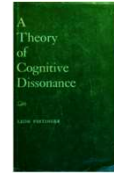
External dissonance

$$H_{ext} = - \sum_i \sum_k \alpha_k s_{ik} b_{ik}$$

Felt dissonances

Depend on **attention** to personal, social, and external dissonances

$$\beta_{pers}, \beta_{soc}, \beta_{ext}$$



Dalege, Galesic, & Olsson, *Psychological Review* (2024)

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Testing model predictions

Surveys: Mechanical Turk, N=973 and USC Understanding America panel (two waves, participants and their friends), N=669

Beliefs

Focal belief: *It is very unsafe or very safe to eat GM food?*

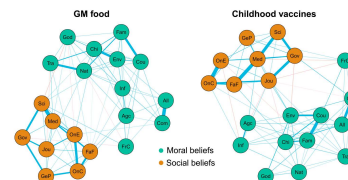
Relevant personal beliefs:

Growing GM food is harmful or beneficial to the environment?

It is very unnatural or very natural to eat GM food?

Social beliefs:

Now think about the opinions of [Contact 1]. Do you think [Contact 1] would say that: It is very unsafe or very safe to eat GM food?



Attention to dissonance

Personal: *It is important to me that my beliefs toward GM food are not in conflict with each other.*

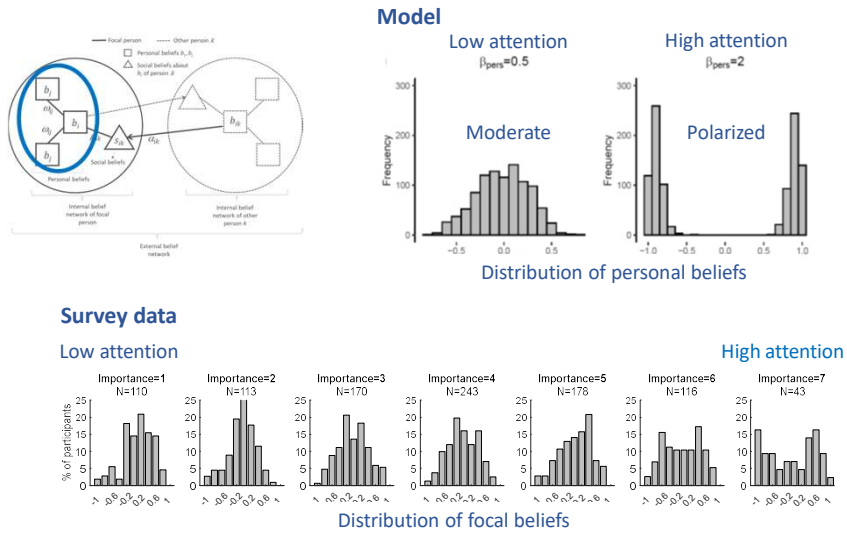
Social: *It is important to me that my personal beliefs and the beliefs of [Contact 1] toward GM food are not in conflict with each other.*

External: *It is important to me that I know what [Contact 1] thinks about GM food.*

→ Factor analysis confirms that those are three different concepts

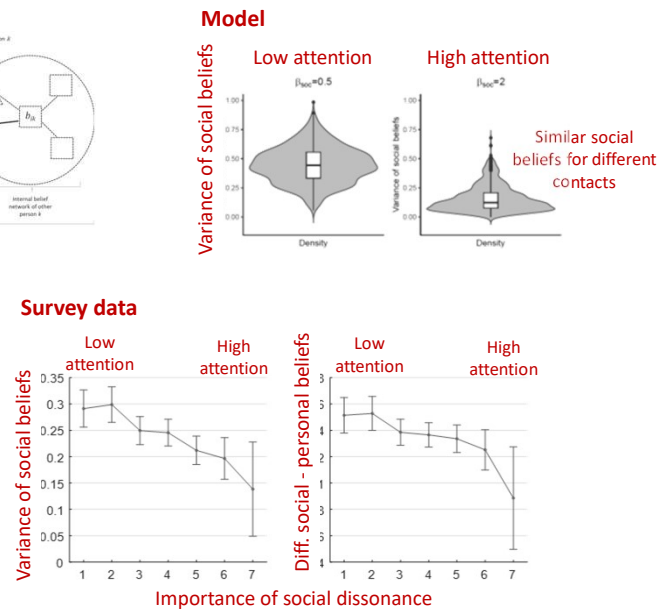
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Effects of attention to Personal dissonance



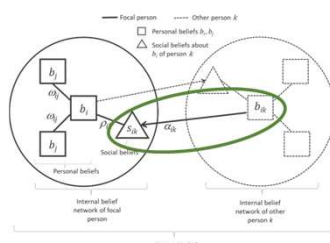
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Effects of attention to Social dissonance

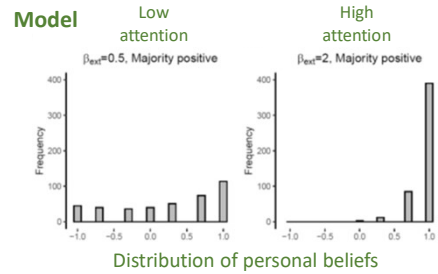


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Effects of attention to External dissonance

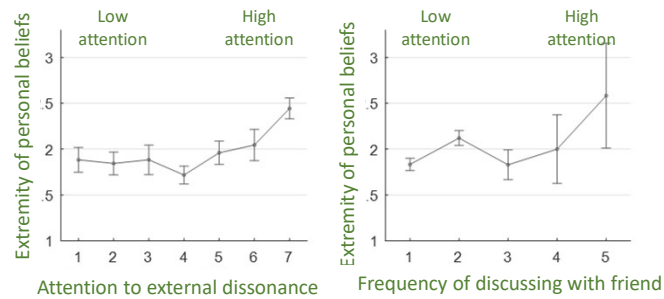


Model



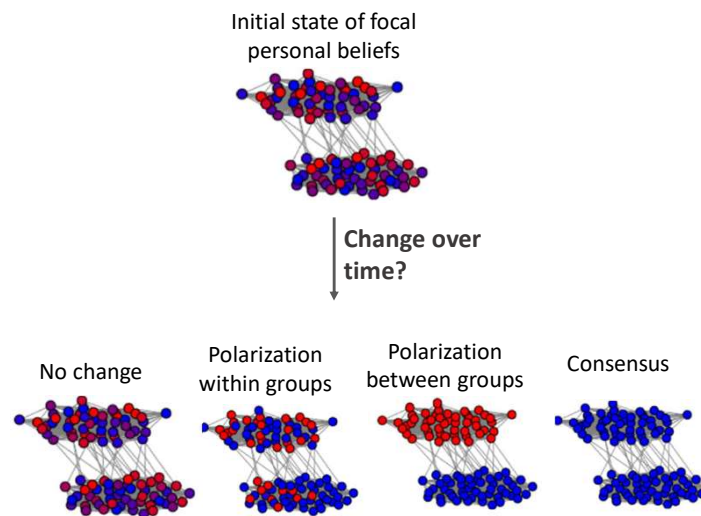
More attention =
more extreme
social beliefs
→ more
extreme
personal
beliefs

Survey data



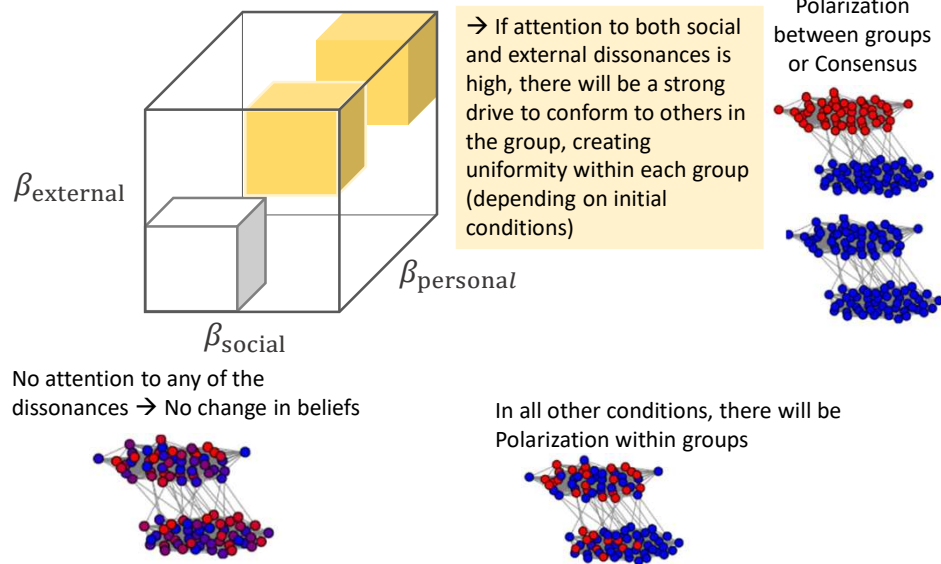
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Explaining polarization and consensus



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Explaining polarization and consensus



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Outlook

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Vision

- Identify the fundamental cognitive and social mechanisms underlying collective adaptation and belief dynamics
- Develop rigorous modeling and empirical paradigms for studying related social phenomena

→ So that we can better understand and predict our collective behaviors and avoid harmful pathways



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We need a new kind of social science

A social science that:

- Can help us understand important social phenomena
- Is transdisciplinary
- Connects rigorous models with rich knowledge of social and cognitive sciences
- Produces quantitative predictions that can be tested empirically
- Helps us to avoid the worst aspects of social dynamics

Your face here



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